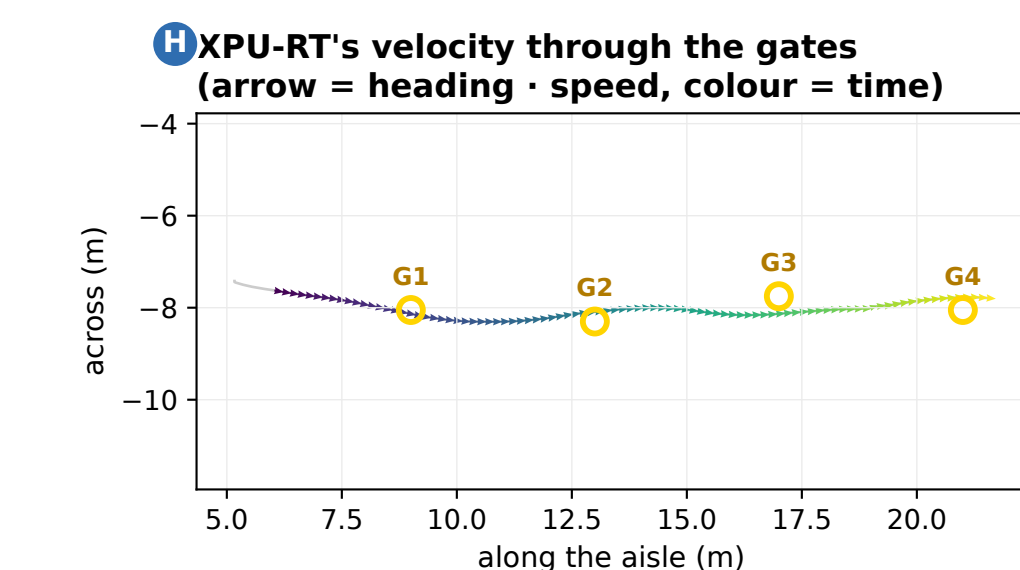
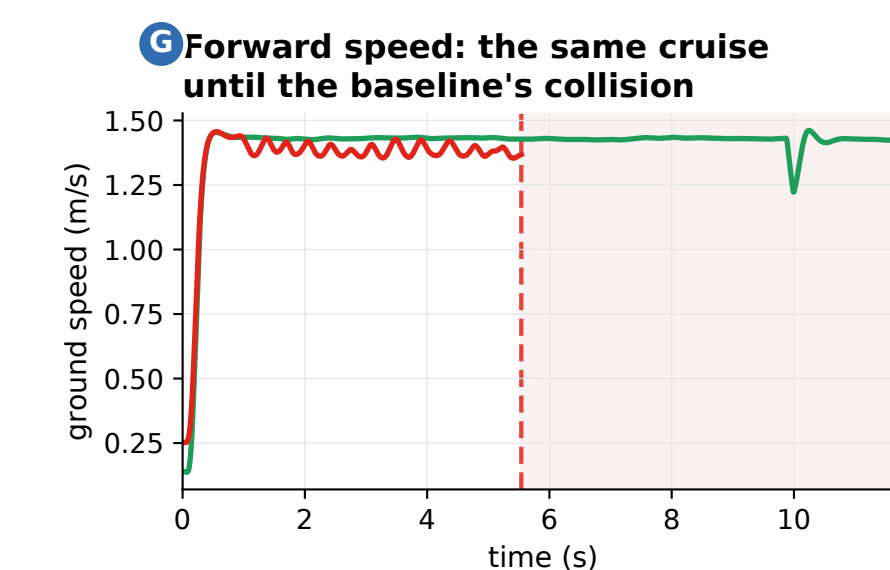
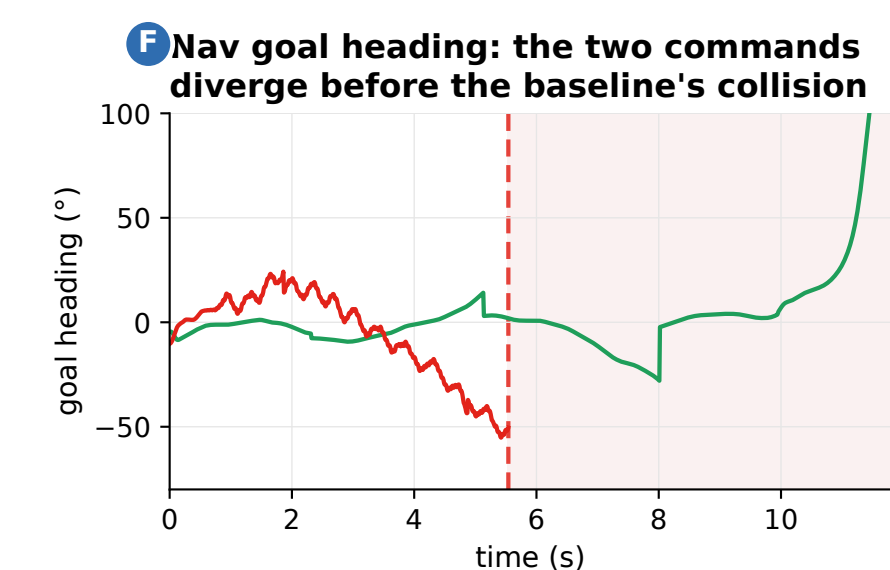
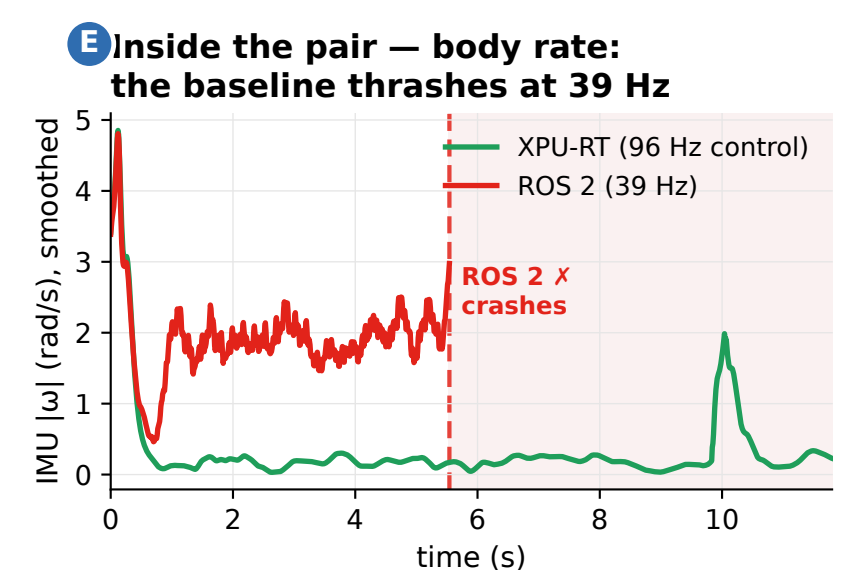
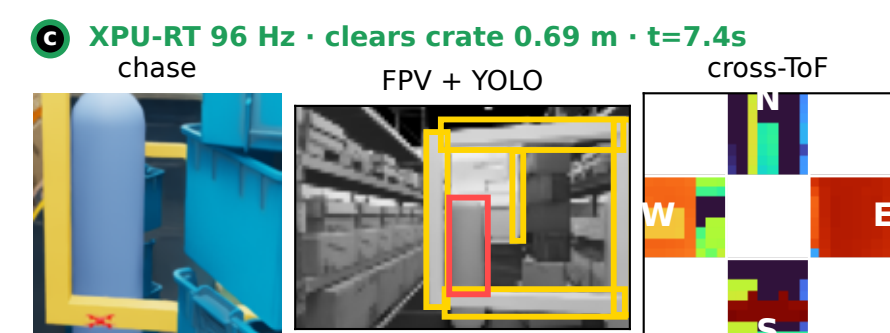
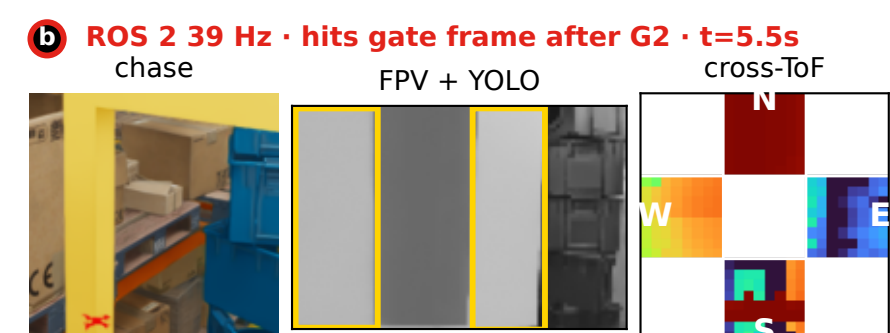
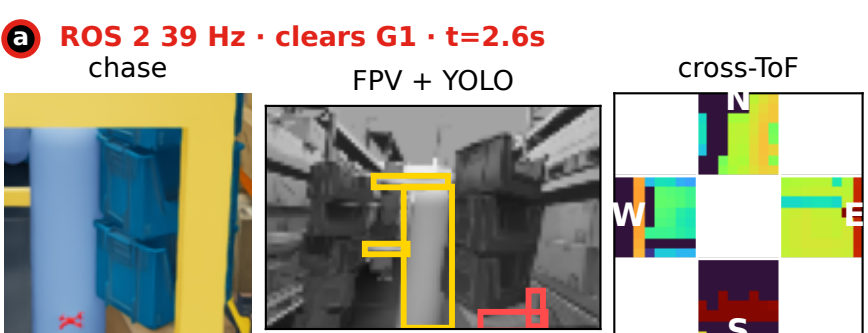
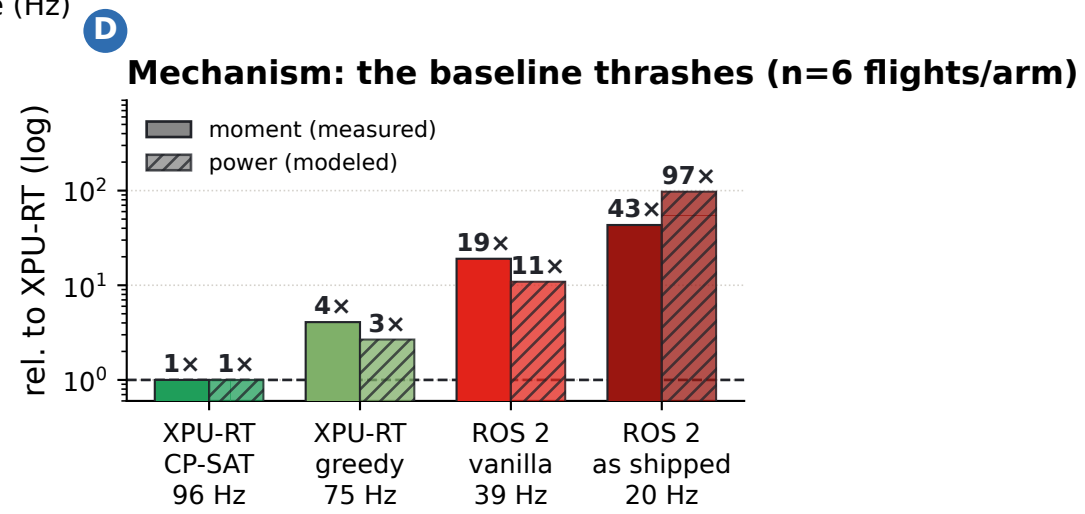
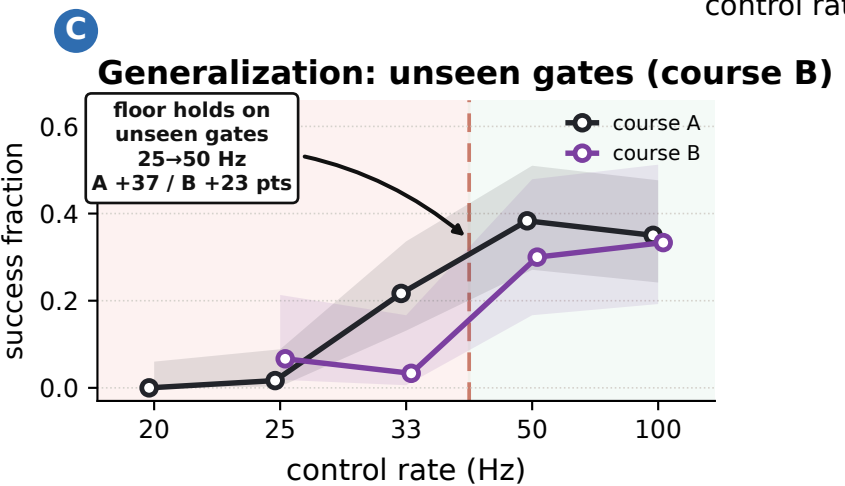
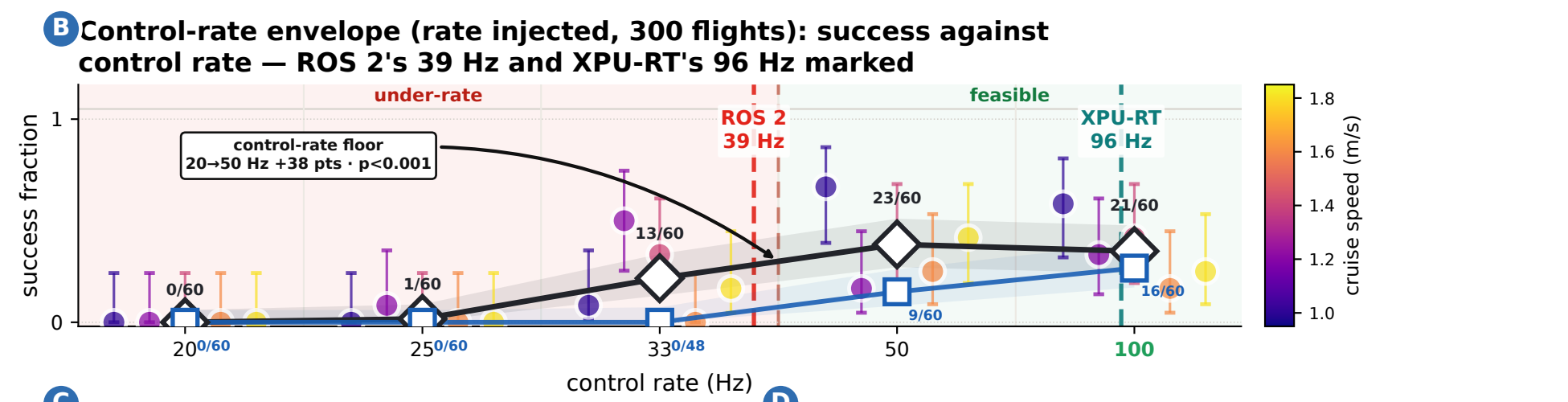
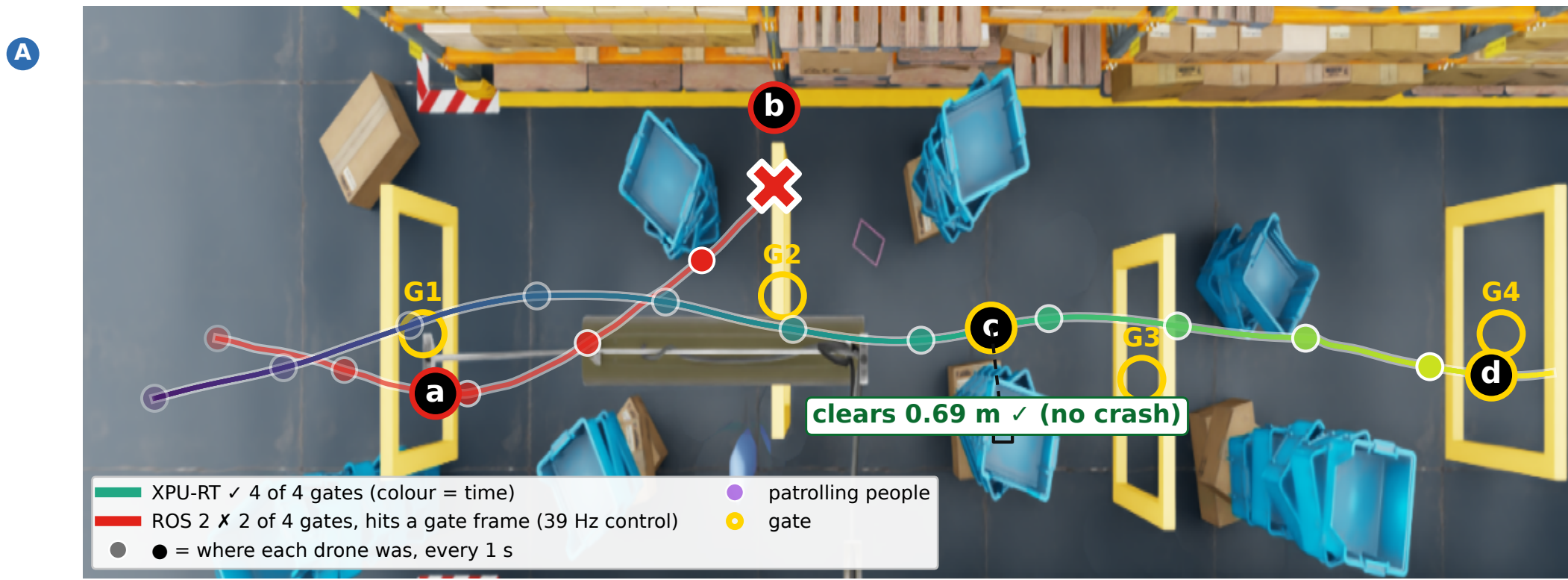


Warehouse gate-course showdown — same scene, drone and gain, at 1.4 m/s
XPU-RT · CP-SAT (96 Hz control) clears all 4 gates
ROS 2 vanilla (39 Hz control) clears G2 → hits a gate frame
scene 1000 · seeds: XPU-RT 1100, ROS 2 1000; over its 12 seeds XPU-RT completes 4, ROS 2 none



I Onboard K1 schedule, measured at the 45 Hz camera — XPU-RT · CP-SAT places the kernels over all 8 harts: camera→control 57 ms, control every 10.0 ms, every frame on time · YOLO on 8 harts, control on 8 harts, nav on 7 harts
ROS 2 (4 processes · single executor · chained control · unpinned): camera→control 242 ms, control every 26.0 ms, 760/766 frames late · YOLO on 4 harts, nav on 1 hart, control on 1 hart

